Vision Transformers for Satellite Image Classification: Zero-Shot, Linear Probe and Fine-Tuning, with a Robustness Study

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Overview

This project applies pre-trained transformer models to a four-class satellite land-cover dataset
(`cloudy`, `desert`, `green_area`, `water`) and asks:

> How well do zero-shot, frozen-feature (linear probe) and fine-tuned vision transformers classify satellite images,
> and do their predictions rely on real scene content or on dataset shortcuts such as image resolution and colour?

Method	Model
Zero-shot classification	CLIP ViT-B/32 (`openai/clip-vit-base-patch32`)
Linear probe on frozen embeddings	CLIP ViT-B/32 and DINOv2-base (`facebook/dinov2-base`)
Fine-tuning	Supervised ViT (`google/vit-base-patch16-224`)
Stress tests / retrieval / statistics	all of the above

Dataset

* Source: **[https://www.kaggle.com/code/sujithmandala/satellite-image-classification-cnn/input]**
* 5,631 images: cloudy 1,500 · desert 1,131 · green_area 1,500 · water 1,500
* **Dataset artefact found:** cloudy and desert images are 256x256, while green_area and water images are 64x64, so
  image size alone predicts the class pair (a size-only rule scores about 53% vs 25% chance). A colour-only baseline
  (mean R, G, B) reaches about 82%. The robustness notebook tests whether the models exploit this.
* The data are **not included** in this repository; see `data/README.md`.
* Split: duplicate-aware (exact and near-duplicate images are grouped), stratified 70/15/15 → 3,942 train / 846 validation /
  843 test images, seed 42. The split is stored in `data/split.csv`.

Notebooks (run in this order)

#	Notebook	Colab	Inputs	Produces	Approx. time (T4)
1	`01_setup_explore_zeroshot_clip.ipynb`: audit, duplicate check, split, zero-shot CLIP	[Open](https://colab.research.google.com/github/muskankh/Satellite-vit-project/blob/main/notebooks/01_setup_explore_zeroshot_clip.ipynb)	`data-2.zip`	`split.csv`, CLIP embeddings, results JSON	~5 min
2	`02_linear_probe_clip_dinov2.ipynb`: linear probes, learning curve	[Open](https://colab.research.google.com/github/muskankh/Satellite-vit-project/blob/main/notebooks/02_linear_probe_clip_dinov2.ipynb)	`split.csv`, `data-2.zip`	embeddings, results JSON	~10 min

| 3 | `03_finetune_vit.ipynb`: fine-tune ViT, LR search, 3 seeds | [Open](https://colab.research.google.com/github/muskankh/Satellite-vit-project/blob/main/notebooks/03_finetune_vit.ipynb) | `split.csv`, `data-2.zip` | saved model, predictions, results JSON | ~20-30 min |
| 4 | `04_robustness_retrieval_stats.ipynb`: robustness, statistics, retrieval, UMAP | [Open](https://colab.research.google.com/github/muskankh/Satellite-vit-project/blob/main/notebooks/04_robustness_retrieval_stats.ipynb) | `split.csv`, `data-2.zip`, model from Notebook 3 | figures, results JSON | ~10-20 min |

How to reproduce

1. Open a notebook in Google Colab (button above) and set **Runtime → Change runtime type → T4 GPU**.
2. Drag `data-2.zip` and `data/split.csv` into the Colab **Files** panel (or keep them in `My Drive/satellite_project/`).
3. Each notebook has a `QUICK_TEST = True` option in its config cell: a small dry run to check that everything works.
   Set it back to `False` for the real run (quick-test numbers are meaningless).
4. Run the notebooks in order. Notebook 4 needs the model saved by Notebook 3 (`vit_best_model/`).
5. All random seeds are fixed (`SEED = 42`; fine-tuning uses seeds 42, 43, 44). Tiny numeric differences can occur
   between GPU types and library versions.

Results summary

Test set = 843 images. 95% confidence intervals come from bootstrap resampling of the test set.

Method	Test accuracy	Macro-F1
Chance	25.0%	n/a
Image-size-only rule (reference)	53.3%	n/a
Colour-only logistic regression (reference)	82.4%	n/a
Zero-shot CLIP ViT-B/32 (prompt: rich class names, "a satellite image taken from space showing …")	74.6% (CI 71.5-77.6)	74.8%
Linear probe: CLIP ViT-B/32 (C = 0.01)	99.88% (CI 99.5-100.0)	99.89%
Linear probe: DINOv2-base (C = 0.01)	99.64% (CI 99.2-100.0)	99.67%
Fine-tuned ViT (mean ± std over 3 seeds; lr 1e-5, 4 epochs)	99.96 ± 0.07%	99.96 ± 0.06%

The three trained methods are statistically indistinguishable on this test set (0-3 errors in 843 images; paired bootstrap and McNemar, Bonferroni-corrected). Only zero-shot CLIP differs (about 25 points lower, $p < 1e-60$). Fine-tuned ViT: seed 42 = 99.88% (CI 99.6-100.0), 3-seed ensemble = 100%.

Robustness: test accuracy (%) when the test images are altered (models not retrained; see `results/results_notebook4.json`

and `results/robustness_drops.csv`). All changes are significant (paired 95% bootstrap CI excludes zero) except zero-shot with same-resolution and the DINOv2 probe when darkened.

Model	Original	Same resolution (all 64x64)	Greyscale	Hue shift	Darker	Brighter
Zero-shot CLIP	74.6	73.5	55.5	40.5	65.2	61.8
CLIP probe	99.9	87.8	77.0	88.6	90.6	89.3
DINOv2 probe	99.6	98.5	88.5	98.3	99.4	99.1
Fine-tuned ViT (seed 42)	99.9	98.2	89.9	98.2	75.8	87.9

Key findings: the CLIP probe loses 12.1 points when the resolution cue is removed (cross-group error 0% to 12.0%), consistent with partial use of resolution/sharpness, while DINOv2 and the fine-tuned ViT lose under 2 points. The fine-tuned ViT is the most fragile to darkening (-24.1 points). Greyscale costs every model 10-23 points. Retrieval: image-to-image precision@1 is 99.2% (CLIP) and 99.5% (DINOv2); CLIP text-to-image retrieval works for clouds, desert and forest but fails for water queries.
Compute for all fine-tuning runs: about 0.27 GPU-hours on a Tesla T4.

Reference baselines (size-only, colour-only) were computed on a separate random split in Notebook 1 and are shown for context.

Repository structure

```
notebooks/   four notebooks (run in order)
data/        split.csv + data description (images are not included)
results/     JSON/CSV result files from each notebook
figures/     figures used in the report
report/      technical report (PDF and Word)
slides/      presentation (PowerPoint)
requirements.txt
```

Trained model

The fine-tuned model (about 330 MB) is too large for GitHub.
Download: **[PASTE GOOGLE DRIVE / HUGGING FACE LINK]**. Place the folder `vit_best_model/` next to the notebooks (or in `My Drive/satellite_project/outputs_nb3/`) before running Notebook 4.

Limitations

* One dataset with four coarse classes and a single train/validation/test split.
* The learning-rate search uses one seed and a small grid.
* Robustness perturbations are synthetic; a drop under a perturbation is evidence of sensitivity, not proof of the cause.
* Image size and class are confounded in the data, so their effects cannot be fully separated without new data.

Ethics and responsible use

Satellite-image classifiers can support environmental monitoring but are dual-use (surveillance). The geographic coverage of the dataset is unknown, so results may not transfer to other regions, sensors or seasons. See the report for discussion.

References

Dosovitskiy et al. (2021) ViT, ICLR · Radford et al. (2021) CLIP, ICML · Oquab et al. (2024) DINOv2, TMLR · Geirhos et al. (2020) Shortcut learning, Nature Machine Intelligence · Loshchilov and Hutter (2019) AdamW, ICLR · Wolf et al. (2020) Transformers, EMNLP Demos · Pedregosa et al. (2011) scikit-learn, JMLR · McInnes et al. (2018) UMAP, arXiv:1802.03426 · McNemar (1947) Psychometrika. Full list in the report.